SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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Register Number	**192311151**

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I sanjay kumar soy(192311151) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: sanjay kumar soy

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Answer

1.A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario

Auto-Scaling and Load Balancing

Auto-scaling automatically increases the number of servers when website traffic rises and reduces them when traffic decreases. This ensures that enough resources are always available to handle customer requests.

Load balancing distributes incoming user requests evenly across multiple servers. If one server becomes overloaded or fails, the load balancer redirects traffic to healthy servers, preventing system failures.

Reducing Downtime and Improving User Experience

By combining auto-scaling and load balancing:

- The website remains available even during peak traffic.
- Users experience faster page loading and minimal delays.
- Server overloads are avoided, reducing downtime.
- Customers can browse products and complete purchases without interruptions, improving overall user satisfaction.

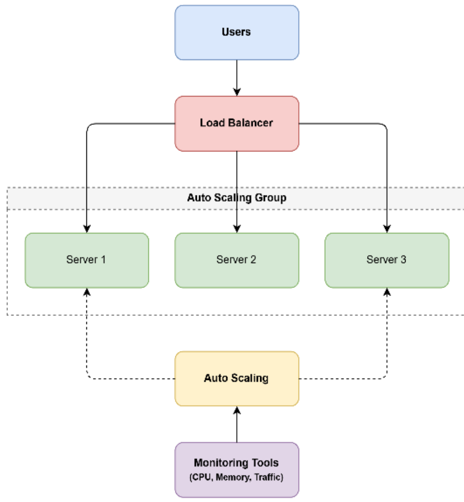
Role of Monitoring Tools

Cloud monitoring tools continuously track system performance, including CPU usage, memory usage, network traffic, and response time. When resource usage reaches a predefined threshold, the monitoring system automatically triggers auto-scaling and sends alerts to administrators if any issues occur. This helps identify problems early and ensures the application runs efficiently.

Real-World E-Commerce Scenario

During the Amazon Great Indian Festival or Flipkart Big Billion Days sale, millions of customers visit the website simultaneously. Auto-scaling automatically launches additional servers to handle the increased traffic, while load balancers distribute customer requests across these servers. Monitoring tools continuously observe system performance and trigger scaling whenever demand increases. As a result, customers can browse products, place orders, and make payments smoothly without website crashes or long waiting times.

Architecture diagram



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1. **Durability and Data Redundancy**

Cloud providers store multiple copies of backup data across different servers and data centers. This is called data redundancy. If one server or storage device fails, the data can be recovered from another copy. This ensures high durability and prevents permanent data loss.

2. Encryption and Access Control

Sensitive backup data is protected using encryption. Data is encrypted both at rest (while stored) and in transit (while being transferred over the network), making it unreadable to unauthorized users.

Access control is implemented using Identity and Access Management (IAM). Only authorized users can access or restore backup data. Features such as Multi-Factor Authentication (MFA) and role-based permissions provide additional security.

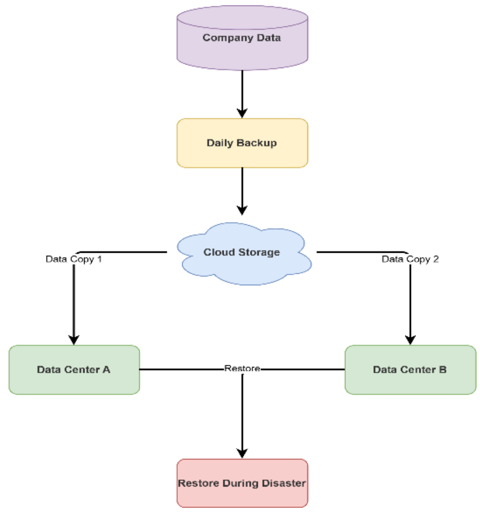
3. Cost Advantages

Compared to traditional storage systems, cloud storage offers several cost benefits:

- No need to purchase and maintain expensive storage hardware.
- Users pay only for the storage they actually use (pay-as-you-go model).
- Maintenance, upgrades, and backups are managed by the cloud provider.
- Storage capacity can be increased or decreased easily without additional infrastructure costs.

4. Disaster Recovery Use Case

A company stores daily backups of its customer database in the cloud. If a fire, flood, or hardware failure damages the on-premises servers, the company can quickly restore the latest backup from the cloud. This minimizes downtime and allows business operations to continue with minimal data loss.



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1. How CDNs Improve Performance and Reduce Latency

A CDN caches images, videos, and other static content on **edge servers** located in multiple geographic locations. When a user requests content, it is delivered from the nearest edge server instead of the main data center. This reduces the distance data must travel, resulting in lower latency, faster loading times, and better application performance.

2. How Edge Servers Enhance User Experience

Edge servers are placed close to users in different regions. They:

- Deliver content more quickly.
- Reduce buffering during video streaming.
- Provide faster downloads and smoother gameplay.
- Reduce the load on the main server, improving overall system efficiency.

This ensures a consistent experience for users regardless of their location.

3. How Cloud Providers Support Global Availability

Cloud providers such as AWS, Microsoft Azure, and Google Cloud have multiple regions and availability **zones** worldwide. They also provide built-in CDN services, load balancing, and data replication. If one region experiences a failure, traffic is automatically redirected to another healthy region, ensuring that the application remains available.

4. Real-Time Streaming or Gaming Example

Consider Netflix or an online multiplayer game like PUBG Mobile. Users from different countries access the service simultaneously. The CDN delivers videos or game updates from the nearest edge server, reducing buffering and lag. Cloud infrastructure automatically handles high traffic and ensures uninterrupted service, providing a smooth viewing or gaming experience.

4.A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Financial institutions handle highly sensitive customer and transaction data. Therefore, they must follow strict security standards and government regulations when using cloud services. Cloud providers offer built-in security and compliance features to help organizations protect their data.

Regulatory Compliance and Auditing

Cloud providers support compliance with standards such as **PCI-DSS, ISO 27001, GDPR, and SOC 2**. They also provide **audit logs** that record all user activities, such as logins, file access, and configuration changes. These logs help organizations verify compliance, detect suspicious activities, and support security audits.

Role of Logging, Monitoring, and Encryption

- **Logging** records all system and user activities, making it easier to trace security events.
- **Monitoring** continuously checks the health and security of cloud resources and generates alerts when unusual activities are detected.
- **Encryption** protects sensitive financial data both **at rest** (stored data) and **in transit** (data being transferred). This ensures that unauthorized users cannot read confidential information.

Together, these security measures protect customer data and reduce the risk of cyberattacks.

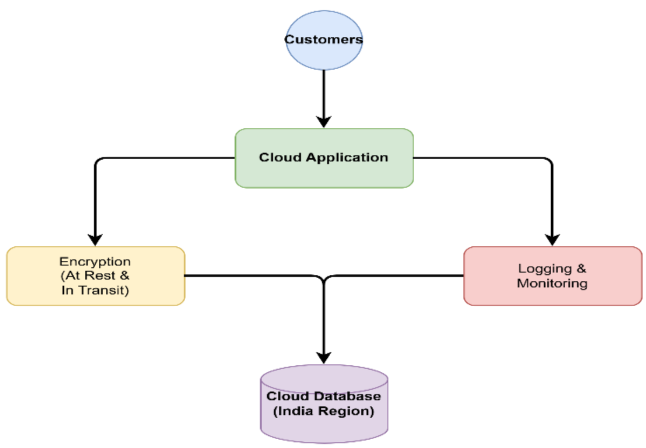
Ensuring Data Sovereignty

Data sovereignty means that data must be stored and processed within a specific country to comply with local laws. The company can meet this requirement by choosing cloud regions located in the required country and ensuring that backups and replicated data also remain within that region.

Example – Financial Transactions

A bank processes online fund transfers through a cloud application. Customer account details and transaction records are encrypted before storage. Every login and transaction is recorded in audit logs, while monitoring tools detect suspicious activities such as unauthorized access attempts. The bank stores all customer data in an Indian cloud region to comply with national data sovereignty regulations.

Cloud providers help financial institutions maintain security, compliance, and regulatory requirements through auditing, logging, monitoring, encryption, and region-specific data storage. These features protect sensitive financial information while ensuring compliance with legal and industry standards.



5.A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability

benefits across different cloud environments. Provide an example of a microservices-based application.

1. Containers vs Virtual Machines

Containers are lighter and faster than Virtual Machines (VMs). They share the host operating system, so they require less memory and storage. As a result, containers start within seconds and use fewer resources.

Virtual Machines include a complete operating system for each instance, making them larger, slower to start, and more resource-intensive. Therefore, containers provide better performance and higher resource utilization.

2. How Kubernetes Simplifies Deployment and Scaling

Kubernetes is a container orchestration platform that automates the deployment, management, and scaling of containers.

It provides:

- Automatic deployment of containerized applications.
- Auto-scaling by increasing or decreasing the number of containers based on traffic.
- Load balancing to distribute requests evenly among containers.
- Self-healing, where failed containers are automatically restarted or replaced.

These features reduce manual effort and improve application reliability.

3. Portability Across Cloud Environments

Containers package the application and all its dependencies together, allowing the same container to run on AWS, Microsoft Azure, Google Cloud, or on-premises servers without modification. This improves portability and avoids vendor lock-in.

4. Example of a Microservices-Based Application

An online shopping application can be built using microservices. Separate containers are used for:

- User Service – Handles user registration and login.
- Product Service – Manages product details.
- Order Service – Processes customer orders.
- Payment Service – Handles online payments.

Each service runs independently in its own container and can be updated or scaled without affecting the others.